

① safirna ploščica, topl. prev. $\lambda_{||}, \lambda_{\perp}, \lambda_1$

v lastnem sist.: $\underline{\lambda}' = \begin{bmatrix} \lambda_{||} & 0 \\ 0 & \lambda_{\perp} \end{bmatrix}$

$a = 5 \text{ mm}$ deb.

$l = 100 \text{ mm}$ šir.

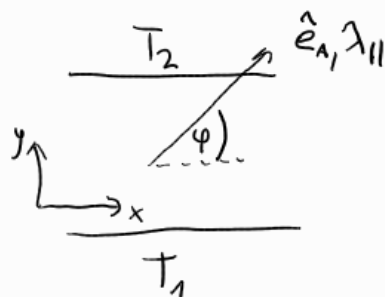
$$\lambda_{||} = 36 \frac{\text{W}}{\text{mK}}$$

$$\lambda_{\perp} = 32 \frac{\text{W}}{\text{mK}}$$

$$\Delta T_x = 10 \text{ K}$$

$$\varphi = 45^\circ$$

$$\underline{j} = ?$$



$$\underline{j} = -\underline{\lambda} \nabla T$$

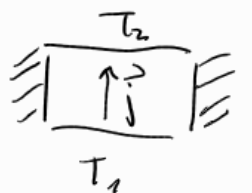
$$\varphi = -45^\circ$$

$$\underline{\lambda} \text{ v } xy \text{ bazi: } \underline{\lambda} = R(\varphi) \underline{\lambda}' R(\varphi)^T =$$

$$= \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_{||} & 0 \\ 0 & \lambda_{\perp} \end{bmatrix} \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\lambda_{||} - s\lambda_{\perp} & 0 \\ s\lambda_{||} + c\lambda_{\perp} & 0 \\ 0 & 0 & \lambda_{\perp} \end{bmatrix} = \begin{bmatrix} c^2\lambda_{||} + s^2\lambda_{\perp} & -sc\lambda_{||} + sc\lambda_{\perp} & 0 \\ -sc\lambda_{||} + sc\lambda_{\perp} & s^2\lambda_{||} + c^2\lambda_{\perp} & 0 \\ 0 & 0 & \lambda_{\perp} \end{bmatrix}$$

a) ploščica je topl. izol. v $\perp$ smeri



$$\Rightarrow j_x = 0 \Rightarrow \underline{j} = j_y \underline{e}_y = -\underline{\lambda} \nabla T = \left[\partial_y T = \frac{\Delta T}{a} \right]$$

$$= - \begin{bmatrix} c^2\lambda_{||} + s^2\lambda_{\perp} & -sc\lambda_{||} + sc\lambda_{\perp} & 0 \\ -sc\lambda_{||} + sc\lambda_{\perp} & s^2\lambda_{||} + c^2\lambda_{\perp} & 0 \\ 0 & 0 & \lambda_{\perp} \end{bmatrix} \begin{bmatrix} \partial_x T \\ \frac{\Delta T}{a} \\ 0 \end{bmatrix} =$$

$$= - \begin{bmatrix} (c^2\lambda_{||} + s^2\lambda_{\perp}) \partial_x T + (-sc\lambda_{||} + sc\lambda_{\perp}) \frac{\Delta T}{a} \\ (-sc\lambda_{||} + sc\lambda_{\perp}) \partial_x T + (s^2\lambda_{||} + c^2\lambda_{\perp}) \frac{\Delta T}{a} \\ 0 \end{bmatrix}$$

$$j_x = 0 \Rightarrow \partial_x T = \frac{-sc\lambda_{||} + sc\lambda_{\perp}}{c^2\lambda_{||} + s^2\lambda_{\perp}} \frac{\Delta T}{a} = - \frac{\lambda_{||} - \lambda_{\perp}}{\lambda_{||} + \lambda_{\perp}} \frac{\Delta T}{a} = - \frac{1}{17} \frac{\Delta T}{a}$$

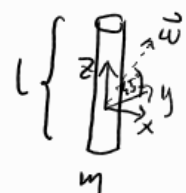
$$j_y = - \frac{\Delta T}{a} \left[\frac{1}{2}\lambda_{||} + \frac{1}{2}\lambda_{\perp} - \frac{\lambda_{||} - \lambda_{\perp}}{\lambda_{||} + \lambda_{\perp}} \left(\frac{1}{2}\lambda_{||} - \frac{1}{2}\lambda_{\perp} \right) \right] = \dots$$

če bi rotirali ploščice, bi (-) pri sin zamenjali. Ker $R(\varphi) = \begin{bmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$ rotiramo bazo, je (-) v sp. vrsti,

b) tok prosta tečje v x smeri $\Rightarrow \partial_x T = 0$

$$\vec{J} = \begin{bmatrix} -sc\lambda_{||} + sc\lambda_{\perp} \\ s^2\lambda_{||} + c^2\lambda_{\perp} \\ 0 \end{bmatrix} \frac{\Delta T}{a}$$

(6.7) Vrtenje palice



$x, y \approx 0$

$$\vec{J} = \int dm \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix} = \int_{-l/2}^{l/2} \frac{m}{l} \begin{bmatrix} z^2 & 0 & 0 \\ 0 & z^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} dz = \frac{m}{l} \begin{bmatrix} \frac{l^3}{12} & 0 & 0 \\ 0 & \frac{l^3}{12} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$dm = m \frac{dz}{l}$$

$$\vec{\omega} = \frac{\omega_0}{\sqrt{2}} (0, 1, 1)$$

$$\left. \frac{z^3}{3} \right|_{-l/2}^{l/2} = \frac{1}{3} \left[\frac{l^3}{8} + \frac{l^3}{8} \right] = \frac{l^3}{12}$$

$$\vec{L} = \int \vec{\omega} = \frac{m l^2 \omega_0}{12\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \underline{\underline{\frac{m l^2 \omega_0}{12\sqrt{2}} (0, 1, 0)}}$$